

1 Basic Knowledge HDE Note

1.1 Linear Algebra

Operations of Matrix Determinant: $\det(kA) = k^n \det(A)$; $\det(A^{-1}) = \det(A)^{-1}$; $(AB)^* = B^*A^*$; $\det(A) = \det(A^T)$.

Eigenvalue λ and Eigenvector $\mathbf{x}$: $A\mathbf{x} = \lambda\mathbf{x}$. 1. Using $\det(A - \lambda I) = 0$ to found λ . 2. Using $(A - \lambda I)\mathbf{x} = 0$ to found $\mathbf{x}$.

· **Algebraic Multiplicity** of an eigenvalue: λ 重复出现的次数. $\geq$ · **Geometric Multiplicity** of an eigenvalue: $\# \mathbf{x}$ under λ .

Generalised Eigenvectors: If λ (AM>GM): Using $\xi_\lambda = (A - \lambda I)\eta_1, \eta_{n-1} = (A - \lambda I)\eta_n$ for each ξ_λ to find all *generalised eigenvectors*.

1.2 ODE / PDE Definitions

Def of DEs / ODEs / PDEs: Given independent variables $(x_1, \dots, x_n)$, functions $(y_1(x_{j_1}), \dots, y_p(x_{j_p}))$ and some derivatives (e.g. $\partial_s y_t$).

· **DEs** if: $\mathcal{F}_i[y_r, \dots, \partial_{x_t} y_b, x_1, \dots, x_n] = 0, \quad i = s, s-1, \dots, 1$. · **ODEs:** $n = 1$ (无偏导) · **PDEs:** $n > 1$ (有偏导)

Def of Linear / Order: Let L be the terms by collecting all the $y_i(x_s)$ (dependent) terms. (e.g. $L = \frac{d^2}{dx^2} - x$ for $y''(x) - xy(x) = x^2$)

· DE is **Linear** if: $L[a_1 y_1 + a_2 y_2] = a_1 L[y_1] + a_2 L[y_2]$ (OR: 所有 $y_i^{(n)}$ 是线性的) · **n-th order ODEs:** $\mathbf{y}^{(n)}(x) = \mathbf{F}(\mathbf{y}_t, \mathbf{y}', \dots, \mathbf{y}^{(n-1)}, x)$

1.3 Calculus & Single ODE / PDE Solution Method

Gradient ∇ : $\nabla F = (\partial_x F, \partial_y F, \partial_z F, \dots)$ **(non-)Simply Connected:** (没)有洞, 相互连接的区域 ∂D : D 区域的边界

Seperable 1st ODE form: $\frac{dy}{dx} = \frac{f(x)}{g(y)}$. $\Rightarrow$ 直接解就行. **Linear 1st ODE form:** $y' + p(x)y = q(x)$. $\Rightarrow e^{\int p(x)dx} y = \int q(x) \cdot e^{\int p(x)dx} dx$

Homogeneous 2nd ODE with constant coefficients: $ay'' + by' + c = 0 \Rightarrow$ Calculate $ar^2 + br + c = 0$ and $\Delta = b^2 - 4ac$.

1. $\Delta > 0, y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$; 2. $\Delta = 0, y = (C_1 + C_2 x) e^{rx}$; 3. $\Delta < 0, y = e^{\alpha x} (C_1 \cos(\beta x) + C_2 \sin(\beta x))$ ps: $r_1, r_2 = \alpha \pm \beta i$

2 Systems of First-Order Linear Equations (ODEs)

2.1 Introduce to First-Order ODEs

* 此节只考虑 First-Order, 所有定理描述都是 First-Order version

n-th ODE $\rightarrow$ 1st ODE: For $y^{(n)} = F(y, y', \dots, y^{(n-1)}, x)$, Let $x_i = y^{(i-1)}$. Then $x'_n = F(x_1, x_2, \dots, x_{n-1})$ which is 1st ODE.

Existence & Uniqueness Theorem: Given $\mathbf{y}'(x) = \mathbf{f}(x, \mathbf{y}_t), \mathbf{y}(x_0) = \mathbf{y}_0$. Let $R_i = [x_0 - t, x_0 + t] \times \prod_i [y_{0_i} - d_i, y_{0_i} + d_i]$.

If $f_j, \partial_{y_i} f_j : R_i \rightarrow \mathbb{R}$ are continuous. $\Rightarrow \exists \delta > 0$ s.t. 1st ODEs has *unique solution* in $[x_0 - \delta, x_0 + \delta]$. In real function

Def of Autonomous: For $\mathbf{y}'(x) = \mathbf{f}(x, \mathbf{y}_t)$. If $\partial_x \mathbf{f} = \mathbf{0} \Rightarrow$ autonomous. 即: $\mathbf{f}(x, \mathbf{y}_t) = \mathbf{f}(x)$ or $\mathbf{f}$ 不含 x

Def of Linear 1st ODEs: $\mathbf{y}'(x) = P(x)\mathbf{y}(x) + \mathbf{g}(x)$, P is matrix-valued function. · **Homogeneous:** $\mathbf{g}(x) = \mathbf{0}$.

· **Existence & Uniqueness theorem** for real linear systems If $P_{ij}(x)$ and $\mathbf{g}(x)$ are continuous on $[\alpha, \beta] \Rightarrow \forall x_0 \in (\alpha, \beta), \mathbf{y}(x_0) = \mathbf{y}_0, \exists! \mathbf{y}(x)$ in $[\alpha, \beta]$

2.2 Homogeneous First-Order Linear Equations Systems

* 此节默认都是 $\mathbf{y}'(x) = P(x)\mathbf{y}(x)$ (*) Solutions $\{\mathbf{y}^{(j)}\}, j = 1, \dots, n$.

Principle of Superposition: If $\mathbf{y}_i$ are solutions $\Rightarrow \sum_i c_i \mathbf{y}_i$ also. · $S := \{\mathbf{y}(x) | \mathbf{y}' = P\mathbf{y}\}$ is a *linear space*, and $\dim(S) = n$ if P_{ij} continuous.

Wronskian at x : $W[\mathbf{y}^{(j)}](x) = \begin{vmatrix} y_1^{(1)}(x) & \dots & y_1^{(n)}(x) \\ \vdots & \ddots & \vdots \\ y_n^{(1)}(x) & \dots & y_n^{(n)}(x) \end{vmatrix}$. · If $W(x) \neq 0 \Rightarrow \{\mathbf{y}^{(j)}\}$ **linearly independent**. ps: 经常用 Abel

· Geometric Meaning: Geometrically, W is the volume spanned by the three vectors $x^{(1)}, x^{(2)}, x^{(3)}$; t 增加或减小代表在 $x^{(i)}$ 方向上 compress/stretch

Fundamental Set of Solutions: If $\{\mathbf{y}^{(j)}(x)\}$ linearly independent at $x \in [\alpha, \beta]$, it is FSoS for that interval.

Fundamental matrix Ψ : If $\{\mathbf{y}^{(j)}(x)\}$ is FSoS. $\Rightarrow \Psi(x) = [\mathbf{y}^{(1)}(x), \dots, \mathbf{y}^{(n)}(x)]$ and $\Psi'(x) = P(x)\Psi(x)$. **Matrix Method**

· $\det[\Psi(x)] = W(x) \neq 0$ · general solution: $\mathbf{y}(x) = \Psi(x)\mathbf{c}, \mathbf{c} = (c_1, \dots, c_n)^T$ · If $\mathbf{y}(x_0) = \mathbf{y}_0 \Rightarrow \mathbf{c} = \Psi^{-1}(x_0)\mathbf{y}_0$

Abel's Theorem: If $W(x_0) \neq 0, x_0 \in I \Rightarrow \forall x \in I, W(x) \neq 0$ P(x) is continuous in I.

Abel's Formula: $W(x) = W(x_0) e^{\int_{x_0}^x \text{tr}[P(s)] ds}; x_0, x \in I$ P(x) is continuous in I.

2.3 Solve $\mathbf{y}' = A\mathbf{y}$

* 此节默认都是 Given constant matrix $A, \mathbf{y}' = A\mathbf{y}$ (*) Calculate λ_j, ξ_j of $A\xi = \lambda\xi$ (找 eigenvalue & eigenvector) Let $T = [\xi_1, \dots, \xi_n], D = \text{diag}(\lambda_1, \dots, \lambda_n)$ (If it's possible)

* Way 1-2 仅 Algebraic Multiplicity = Geometric Multiplicity

Solve $\mathbf{y}' = A\mathbf{y}$ (way 1): $\mathbf{y} = \sum_j c_j e^{\lambda_j x} \xi_j$ (general solution)

Solve $\mathbf{y}' = A\mathbf{y}$ (way 2): 接着 1. If $\exists(\lambda, \bar{\lambda}) \in \mathbb{C} \Rightarrow e^{\lambda t} \xi = \mathbf{u} + \mathbf{v}i \Rightarrow$ Corresponding sol: $c_1 \mathbf{u} + c_2 \mathbf{v} + \dots$

Solve $\mathbf{y}' = A\mathbf{y}$ (way 3): $\mathbf{y}(x) = e^{Ax} \mathbf{y}(0), e^{Ax} = T e^{Dx} T^{-1}$ (e^{Ax} 是 **Fundamental matrix**)

Matrix Method or $\mathbf{y}(x) = T\mathbf{z}(x) \Rightarrow [\mathbf{y}' = A\mathbf{y} \Leftrightarrow \mathbf{z}' = D\mathbf{z}]$ with $\mathbf{z} = e^{Dx} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})$

ps **1.** $e^{Ax} = \Psi(x)$ with $\Psi(0) = \mathbb{I}$ **2.** $e^{Ax} = \Psi(x)\Psi^{-1}(0)$

Exponential of a matrix: Given $P = A$ constant matrix, then $e^{Ax} := \sum_{n=0}^{\infty} \frac{(Ax)^n}{n!} = \mathbb{I} + Ax + \frac{1}{2!} A^2 x^2 + \dots, = \lim_{n \rightarrow \infty} (\mathbb{I} + \frac{Ax}{n})^n$

· Let A, B be matrix. Then **1.** $e^{A+B} = e^A e^B$ if $AB = BA$ **2.** $e^{A \cdot 0} = \mathbb{I}$ **3.** $\frac{d(e^{Ax})}{dx} = A e^{Ax}$ 和上面 solve $\mathbf{y}' = A\mathbf{y}$ 相关

Solve $\mathbf{y}' = A\mathbf{y}$ (way 4): 接着 1. (For each ξ_λ , under λ) $\mathbf{y}^{(1)} = e^{\lambda x} \xi_\lambda$ $\mathbf{y}^{(2)} = e^{\lambda x} (x \xi_\lambda + \eta_1)$ $\mathbf{y}^{(3)} = e^{\lambda x} (\frac{x^2}{2!} \xi_\lambda + x \eta_2 + \eta_1), \dots, \mathbf{y}^{(n)} = e^{\lambda x} [\frac{x^{n-1}}{(n-1)!} \xi_\lambda + \sum_{i=0}^{n-2} \frac{x^i}{(i)!} \eta_{i+1}]$

or 接着 3 Let $T = [\lambda_1, \dots, \lambda_i, \eta_1^{(i)}, \dots, \lambda_j, \eta_1^{(j)}, \dots]$. $J = T^{-1}AT$ (Jordan form) $\mathbf{y}' = T\mathbf{z} \Rightarrow [\mathbf{y}' = A\mathbf{y} \Leftrightarrow \mathbf{z}' = J\mathbf{z}]$, easy to solve $\mathbf{z}' = J\mathbf{z}$. ps: e^{Jt} 见上图

2.4 Solve $\mathbf{y}' = A\mathbf{y} + \mathbf{g}$

* 此节默认都是 Given constant matrix $A, \mathbf{y}' = A\mathbf{y} + \mathbf{g}$ (**) 对于 **Variation of Parameters** 方法, A 可以为 P (任意矩阵函数)

Calculate λ_j, ξ_j of $A\xi = \lambda\xi$ Let $T = [\xi_1, \dots, \xi_n], D = \text{diag}(\lambda_1, \dots, \lambda_n)$ (If it's possible, or to Jordan form)

* 此节中, 给出的都是如果是关于 initial value 的解时: 要 general solution 时, $\mathbf{y}_g$ 部分有关 initial value 的部分可以用 $\mathbf{c}$ 代替, $\mathbf{y}_p$ 部分有关 initial value 部分时, 可以任意取值

Theorem: General solution of (**) is of the form: $\mathbf{y} = \sum_i c_i \mathbf{y}^{(i)} + \mathbf{y}_p$, where $\mathbf{y}^{(i)}$ is solution of (*). 通常会让特解中积分后的常数 $\mathbf{c} = \mathbf{0}$

Variation of Parameters: $\mathbf{y} = \mathbf{y}_g + \mathbf{y}_p$; $\mathbf{y}_g = \Psi(x)\Psi^{-1}(x_0)\mathbf{y}(x_0)$; $\mathbf{y}_p = \Psi(x)\mathbf{u}(x), \mathbf{u}(x) = \int_{x_0}^x \Psi^{-1}(s)\mathbf{g}(s)ds$. 经常 $\Psi(x)\mathbf{u}' = \mathbf{g}$ 来找 $\mathbf{u}$

Diagonalisation (matrix methods): ¹ Let $\mathbf{y} = T\mathbf{z}$ ² (**) $\Rightarrow \mathbf{z}' = D\mathbf{z} + T^{-1}\mathbf{g} = D\mathbf{z} + \mathbf{h}$ ³ 直接计算; $\mathbf{z}_i = c_i e^{r_i x} + e^{r_i x} \int_{x_0}^x e^{-r_i s} h_i(s) ds$ $\mathbf{y}_p$

or, 由上面 Way 4 得到 T, J . ¹ Let $\mathbf{y} = T\mathbf{z}$ ² (**) $\Rightarrow \mathbf{z}' = J\mathbf{z} + T^{-1}\mathbf{g} = J\mathbf{z} + \mathbf{h}$ ³ 直接计算; 可能会遇到 $\mathbf{y}' + p(x)\mathbf{y} = q(x)$ 的形式

Undetermined Coefficients: when $\mathbf{g}(x)$ is polynomials, or real/complex exponential. Let $P_n(x) = \sum_{i=0}^n \mathbf{p}_i x^i, A_n(x) = \sum_{i=0}^n \mathbf{a}_i x^i$

¹ 构造 $\mathbf{y}_p$ 与 $\mathbf{g}$ 结构相匹配 (e.g. If $\mathbf{g} = P_n(x) \Rightarrow \mathbf{y}_p = A_n(x)$; If $\mathbf{g} = \mathbf{c} e^{kx} \Rightarrow \mathbf{y}_p = \mathbf{a} e^{kx}$) ² 特殊: If $\mathbf{g} = \mathbf{c} e^{\lambda x}, \lambda$ is eigenvalue with $AM = n \Rightarrow \mathbf{y}_p = e^{\lambda x} \sum_{i=0}^n \mathbf{a}_i x^i$ ³ 带入原方程得未知系数

ξ_i, η_i 默认是向量 | $\mathbf{v}_t$ 代表转置, 如果后面也是就不再重复写 | Copyright By Jingren Zhou | Page 1

2.5 Geometrical Interpretation & Linear Approximation

* 此节中，Consider nonlinear autonomous system: $x' = F(x,y), y' = G(x,y)$ (**) **Phase Plane:** O_{xy} referred as phase plane. **Trajectories:** Sol of ODE. $\mathbf{x}(t) = (x(t), y(t))^T$ direction $(dx/dt, dy/dt)$ **Phases Portrait:** 一组有代表性的轨迹 **Critical Point (Equilibrium Point):** $\mathbf{x}_0 = (x_0, y_0)$ is critical point iff $[x' = F(\mathbf{x}_0) = 0, y' = G(\mathbf{x}_0) = 0]$ or $[\mathbf{x}' = \mathbf{0}]$ (critical point 外变化简单) **Stable:** Critical Point $\mathbf{x}_0$ is stable iff $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall$ sols $\mathbf{x}(t)$ with $||\mathbf{x}(0) - \mathbf{x}_0|| < \delta$, then $||\mathbf{x}(t) - \mathbf{x}_0|| < \varepsilon, \forall t \geq 0$ · 直观理解: 一开始离 critical point 很近, 那么随着时间推移也会保持在 critical point 附近. ps:attracting 是会趋近于 critical point.

Attracting: Critical point $\mathbf{x}_0$ is attracting iff $\exists \delta > 0$ s.t. $\forall$ sols $\mathbf{x}(t)$ with $||\mathbf{x}(0) - \mathbf{x}_0|| < \delta$ satisfies: $\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{x}_0$

Asymptotically Stable: Critical point $\mathbf{x}_0$ is asy.stab iff ¹ Stable, ² Attracting.

Theorem: If $\mathbf{x}_*(t) = (x_*, y_*)$ s.t. $F(\mathbf{x}_*), G(\mathbf{x}_*) \neq 0. \Rightarrow \exists H$ s.t. $H(x,y) = (\tilde{x}, \tilde{y}), \tilde{x}' = 1, \tilde{y}' = 0$ H is a smooth change of variable function in neighbourhood of $\mathbf{x}_*$

Non-Linear $\rightarrow$ Linear: For system (**), assume $\mathbf{x}_0 = (x_0, y_0)$ is critical point. $\Rightarrow \mathbf{u} = \mathbf{x} - \mathbf{x}_0$, then $\mathbf{u}' \approx \mathbf{A}\mathbf{u}$ Linear; $\mathbf{A} = \begin{pmatrix} \partial_x F(\mathbf{x}_0) & \partial_y F(\mathbf{x}_0) \\ \partial_x G(\mathbf{x}_0) & \partial_y G(\mathbf{x}_0) \end{pmatrix}$ Jacobian

2.6 Classification of Critical Points

* 此节中，Consider linear system: $\mathbf{x}' = \mathbf{A}\mathbf{x}$. Critical Point(s): when $\mathbf{x}' = \mathbf{A}\mathbf{x} = \mathbf{0}$

Classification of Critical Points at (x_0, y_0) (Linear): Let r_1, r_2 be sol of $\det(\mathbf{A} - \lambda \mathbf{I}) = 0.$ · $\mathbb{C} : r = \lambda \pm i\mu (\mu > 0)$ · If $\mathbf{A}$ constant, write sol: $\mathbf{x} = c_1 e^{r_1 t} \xi_1 + c_2 e^{r_2 t} \xi_2$ · $GM = 1: \mathbf{x} = c_1 e^{rt} \xi + c_2 e^{rt} (t\xi + \eta)$ · AL: Almost Linear Systems.

$\mathbb{R}/\mathbb{C}$	Condition Stability	Type Name	Phase Plane Description	Other
$\mathbb{R}$	$r_1 < r_2 < 0$ asy.stab	N NSk	向原点, ξ_2 直线, ξ_1 曲线, and ξ_1 周围 $y = \pm x^3$	$c_2 \neq 0, t \rightarrow \infty: \xi_2$ 主导方向; $c_2 = 0, t \rightarrow \infty: \xi_1$ 主导方向
	$r_1 > r_2 > 0$ unstable	N NSo	原点向外, ξ_2 直线, ξ_1 曲线, and ξ_1 周围 $y = \pm x^3$	$c_1 \neq 0, t \rightarrow \infty: \xi_1$ 主导方向; $c_1 = 0, t \rightarrow \infty: \xi_2$ 主导方向
	$r_1 > 0 > r_2$ unstable	SP SP	$t \rightarrow \infty, \xi_1$ 从原点向外, ξ_2 从外向原点 and: 像 $y = \pm \frac{1}{x}$, 同进同出	$t \rightarrow \pm \infty: \mathbf{x} \rightarrow \infty; \quad t \rightarrow \infty: c_1, c_2 \neq 0, \mathbf{x} \rightarrow \infty, \xi_1$ 主导; $t \rightarrow \infty: c_2 = 0, \mathbf{x} \rightarrow \infty, \xi_1$ 主导; $t \rightarrow \infty: c_1 = 0, \mathbf{x} \rightarrow 0, \xi_2$ 主导
	$r_1 = r_2 < 0$, GM=2 asy.stab	PN PN or Stable Star	直线 向原点	直线, u_1/u_2 is t independent
	$r_1 = r_2 > 0$, GM=2 unstable	PN PN or Unstable Star	直线 从原点向外	直线, u_1/u_2 is t independent
	$r_1 = r_2 < 0$, GM=1 asy.stab	IN (AL>Type: SpP) IN (Stable)	S 曲线, 向原点	$t \rightarrow \infty, \mathbf{x} \rightarrow 0, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2 \xi$ 方向相同
	$r_1 = r_2 > 0$, GM=1 unstable	IN (AL>Type: SpP) IN (Unstable)	S 曲线, 从原点向外	$t \rightarrow \infty, \mathbf{x} \rightarrow \infty, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2 \xi$ 方向相同
$\mathbb{C}$	$\lambda \neq 0, \lambda > 0$ unstable	SpP Unstable Focus	向外椭圆 (elliptical) 螺旋	$t \rightarrow \infty, \mathbf{x} \rightarrow \infty$ ps: 考虑 $A = (a, b; c, d)$, 如果 bc>0, 顺时针, 如果 bc<0, 逆时针
	$\lambda \neq 0, \lambda < 0$ asy.stab	SpP Stable Focus	向内椭圆 (elliptical) 螺旋	$t \rightarrow \infty, \mathbf{x} \rightarrow 0$ ps: 考虑 $A = (a, b; c, d)$, 如果 bc>0, 顺时针, 如果 bc<0, 逆时针
	$\lambda = 0$ stable (AL:Indeterminate)	C (AL:C or SpP) C	椭圆 (elliptical) 和 半长轴 ξ 实部方向	Bounded trajectory or $\exists$ Periodic Trajectories
ps: N = Node PN = Proper Node IN = Improper / Degenerate Node SP = Saddle Point SpP = spiral (focus) point C = Center NSk = Nodal Sink NSo = Nodal Source				
For Almost Linear systems (见 2.5): $\mathbf{u}' = \mathbf{A}\mathbf{u} + \eta$, 如果 linear Approximation (去掉 η), 按照 Linear 考虑. 如果不去 η , 按照去掉的算 r_1, r_2 , Type 按照正常 / (AL) 的来.(图像会到 (0,0))				

Some Defs: *Separatrix:* 将不同区域分开的轨迹; *Basin of attraction:* 所有趋近 critical point 的轨迹. ps: In any nonlinear ODEs, it's important to determine the basin of attraction.

2.7 Non-Linear Methods

* 此节中，Consider two-dimensional autonomous system: $x' = F(x,y), y' = G(x,y)$ and $\mathbf{x} = (x,y)^T$ (†) Lyapunov's: 不需要知道微分方程的确切解 critical value. Poincaré-Bendixson: 寻找 limit cycle

Positive (Negative) [semi-]definite: Function $E : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is positive (negative) [semi-]definite iff: ps: positive [semi-]definite 正(半)定 ¹ $E(\mathbf{x}_0) = 0, \forall \mathbf{x} \in D, \mathbf{x} \neq \mathbf{x}_0, E(x,y) > 0$ or $(E(x,y) < 0)$ or $[E(x,y) \geq 0]$ ps: Neighbourhood 领域

Isolated Critical Point: Critical Point $\mathbf{x}_0$ isolated iff $\exists$ Neighbourhood containing no other critical points. $\Leftrightarrow E$:Lyapunov function

Lyapunov Stability Theorem: ¹(#); ²ICP $\mathbf{x}_0 \in D$; ³ $\exists$ 正定且连续 $E : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, E(\mathbf{x}_0) = 0; || \frac{dE}{dt} = \nabla E \cdot \mathbf{x}' = \partial_x E \cdot F + \partial_y E \cdot G$ 负(半)定 on $D \setminus \mathbf{x}_0(D) \Rightarrow \mathbf{x}_0$ asp.(stab)

· **Look for function E :** Way 1: May be energy, but not always. Way 2: e.g. $E = ax^{2n} + by^{2m}, E = a(x+y)^{2n} + by^{2m}$ ps: $\frac{dE}{dt} = \nabla E \cdot \mathbf{x}'$

Lyapunov Instability Theorem: ¹(#); ²ICP $\mathbf{x}_0 \in D$; ³ $\exists E : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, \partial_i E$ 连续, $E(\mathbf{x}_0) = 0; \frac{dE}{dt}$ 正定 on $D \Rightarrow \mathbf{x}_0$ unstable

Periodic Solution: Soln $\mathbf{x}$ has: $\mathbf{x}(t+T) = \mathbf{x}(t), \forall x.$ || 闭合曲线 $\Leftrightarrow$ Periodic sol || Periodic Solution + 附近有吸引或排斥 $\Rightarrow$ Limit Cycle || when $\dot{r} = 0, \dot{\theta} = const.$ Periodic sol

Limit cycle: Limit cycle $\Rightarrow$ Periodic Solution || Def $t \rightarrow \pm \infty, \exists$ non-closed trajectory asymptotes to it. (inside or outside) ps: It's Orbital Stability

- Limit cycle is: ¹asy.stab: $\forall$ trajectories asy. it; ²semistable: $\exists$ trajectories 单方向进/出; ³unstable: $\forall$ trajectories away from it.
- Theorem (If $\exists$ limit cycle / closed):** ¹ F, G 一阶偏导连续 in simply connected D ; ² $\exists$ 极限环/封闭环 $(\star)$ in $D. \Rightarrow \supseteq$ CP in $(\star)$; ²If $\exists!$ CP in $(\star)$, it's not SP.
- Theorem (No limit cycle / closed):** ¹ F, G 一阶偏导连续 in simply connected D ; ² $\partial_x F + \partial_y G$ 符号不变 on $D \Rightarrow$ No closed trajectory in D .

2 的推论: ¹If simple connected D no CP $\Rightarrow$ No closed trajectory in $D.$ ²If simple connected D has unique CP $\Rightarrow$ No closed trajectory in D

Poincaré-Bendixson Theorem: ¹ F, G 一阶偏导连续 in D ; ² $\exists D_1 \subset D$ bounded; ³ $R = D_1 \cup \partial D_1$ has NO CP; || ⁴ $\exists$ soln $\mathbf{x} \in R, \forall t \geq t_0 \Rightarrow \supseteq$ closed trajectory. ² Either $\mathbf{x}$ sprial towards a Periodic trajectory. or $\mathbf{x}$ periodic ³ R is not simply connected

· **Find Trapping Region:** trapping region: 符合上面定理 2-4 除 CP 的 region. Method: Consider polar coordinate, 计算 $r^2 = x^2 + y^2$ 对 t 的求导, 寻找 $r\dot{r} \geq 0$ or $r\dot{r} \leq 0$

3 Fourier Series

* BVPs 可能没有解/一个解/无数解; IVP 只会会有一个解

3.1 Function Space, Eigenfunctions, Periodic Function and Orthogonal Function

Differential Operators: Any linear combination of n -th derivatives defines a linear map or linear operator. e.g. $p(x) \frac{d}{dx} + \frac{d^2}{dx^2} : y \mapsto p(x) \frac{dy}{dx} + \frac{d^2 y}{dx^2}$

Eigenvalues & Eigenfunctions: If D is a differential operator $\Rightarrow$ If $Df = \lambda f$, then λ is Eigenvalue and f is Eigenfunctions

Periodic Functions: Function $f(x)$ is Periodic iff $f(x+T) = f(x), \forall x.$ **Period:** T. **Fundamental Period:** The smallest possible T.

· Property: If f, g has same period $T. \Rightarrow af(x) + bg(x)$ also has period $T.$ || For $\sin(\omega x), \cos(\omega x): T = 2\pi/\omega$

Inner Product of functions: func f, g in the interval $[a, b] \Rightarrow$ Inner Product: $(f, g) = \int_a^b f(x)g(x)dx$ || **Orthogonal:** $(f, g) = 0$

· Important Situation: Set $\{\sin(\frac{n\pi x}{L}), \cos(\frac{m\pi x}{L})\}$ Orthogonal each pair (**Mutually orthogonal**) in the interval $[-L, L].$

Kronecker delta: $\delta_{mn} = \{0, m \neq n; 1, m = n.$

3.2 Fourier Series

Fourier Series: Periodic function with $T = L$, $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L}) \right)$ (*) where a_0, a_n, b_n are determined by Euler-Fourier Formulas.

Euler-Fourier Formulas: Suppose (*) converges in the interval $[-L, L]$. Then:

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos(\frac{n\pi x}{L}) dx \qquad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin(\frac{n\pi x}{L}) dx \qquad a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

4 Appendix

Trigonometric func: s+s=sc, s-s=cs, c+c=cc, c-c=-ss || e.g. $\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2})$ $\sin A \sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$

hyperbolic func: $\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$ $\cosh^2(x) - \sinh^2(x) = 1$ $[\sinh(x)]' = \cosh(x)$ $[\cosh(x)]' = \sinh(x)$

Common Int: $\int \tan(x) dx = -\ln|\cos(x)|$; $\int \cot(x) dx = \ln|\sin(x)|$; $\int \sec(x) dx = \ln|\sec(x) + \tan(x)|$; $\int \csc(x) dx = \ln|\csc(x) - \cot(x)| = \ln|\tan(\frac{x}{2})|$

Common Der: $[\tan(x)]' = \sec^2(x)$; $[\cot(x)]' = -\csc^2(x)$; $[\sec(x)]' = \sec(x)\tan(x)$; $[\csc(x)]' = -\csc(x)\cot(x)$; $[\arctan(x)]' = \frac{1}{1+x^2}$.

$\cdot 1 + \tan^2(x) = \sec^2(x)$; $1 - \csc^2(x) = \cot^2(x)$; $\csc^2(x) - \cot^2(x) = 1$. $[\arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$; $[\arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

Polar Coordinate: $x = r\cos(\theta), y = r\sin(\theta)$ we have: $r\dot{r} = x\dot{x} + y\dot{y}$, $y\dot{x} - x\dot{y} = -r^2\dot{\theta}$